Practice workbook · State the function of a skeletal or joint part and explain effects of damage

Roles of ligaments, tendons, cartilage, synovial fluid, bone tissues and the axial skeleton, and what happens when cartilage or collagen is lost.

13 question parts: 1 worked-example retry and 12 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: HBI-SKELETAL-PART-FUNCTION

1. Question 2 · part b.ii · 4 marks

June 2023 Paper 1R

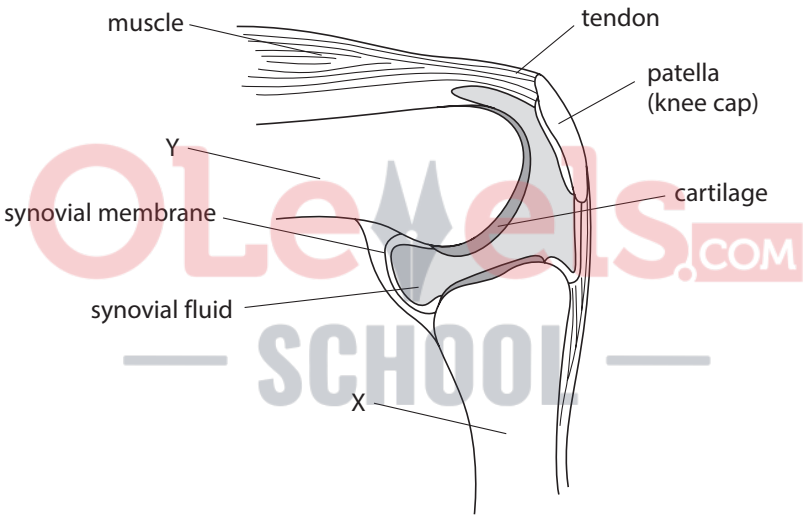
Worked-example retry · Attempt part b.ii only.

2 The skeleton contains joints.

(a) Describe what is meant by the term **joint**.

(2)

(b) The diagram shows a knee joint.



(i) Identify bone X and bone Y.

(2)

X .....

Y .....

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1. Question 2 · part b.ii · 4 marks

June 2023 Paper 1R

Worked-example retry · Attempt part b.ii only.

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(ii) Give the functions of tendons and cartilage.

(4)

tendons

.....

.....

cartilage

.....

.....

(iii) Give two structures, not shown on the diagram, that would be necessary for the joint to fully function.

(2)

1 .....

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(Total for Question 2 = 10 marks)

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1. Question 2 · part b.ii · 4 marks · Working and answer

Working and answer

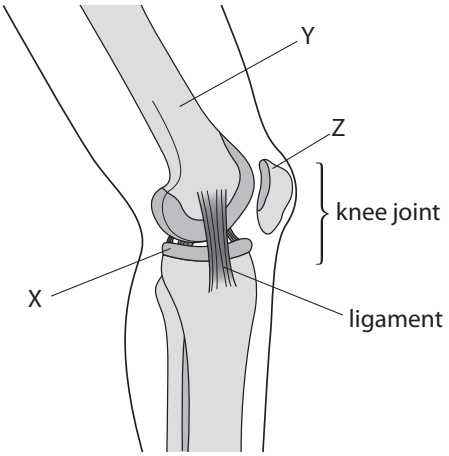
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2. Question 1 · part a.ii · 1 mark

June 2019 Paper 1

Further practice · Attempt part a.ii only.

1 (a) The diagram shows structures in part of a human leg.



(i) The box lists structures in the human leg.

cartilage	femur	kneecap	muscle
fibula	tendon	tibia	

Use words from the box to name structures X, Y, and Z.

(3)

X .....

Y .....

Z .....

(ii) What is the function of ligaments in the knee joint?

(1)

- ☒ **A** ligaments attach muscles to the leg bones
- ☒ **B** ligaments hold the leg bones together
- ☒ **C** ligaments move the leg bones
- ☒ **D** ligaments stop the leg bones rubbing together

(iii) The type of joint found in the knee and the elbow is the same.

Give the name of this type of joint.

(1)

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2. Question 1 · part a.ii · 1 mark · Working and answer

Working and answer

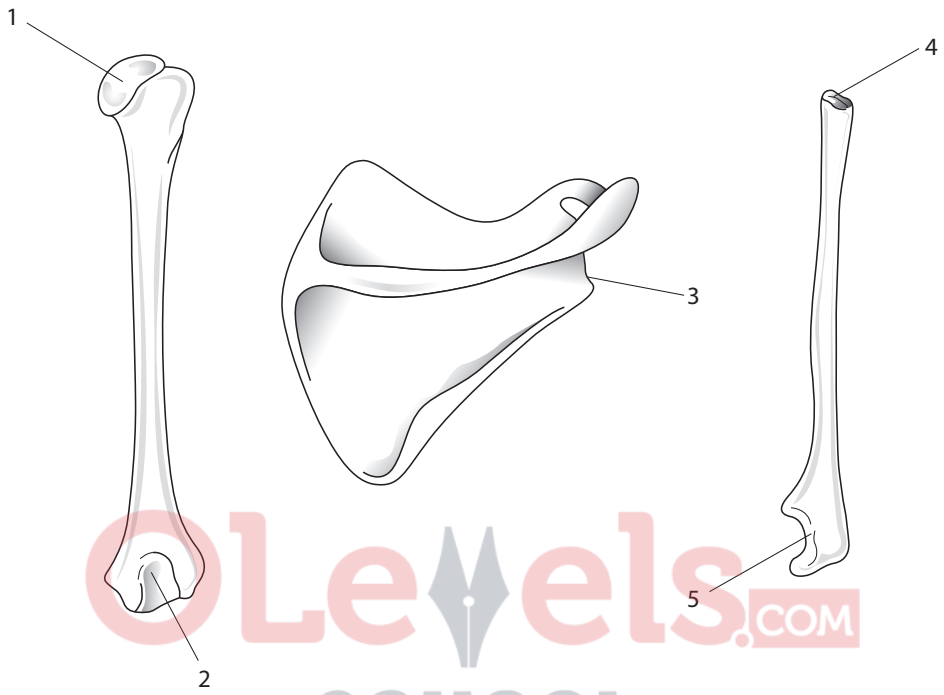
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3. Question 2 · part b.iv · 2 marks

June 2019 Paper 2R

Further practice · Attempt part b.iv only.

2 (a) The diagrams show three bones from the human skeleton.



Which two numbered regions come together to form a ball and socket joint?

(1)

- ☒ **A** 1 and 3
- ☒ **B** 1 and 5
- ☒ **C** 2 and 3
- ☒ **D** 2 and 4

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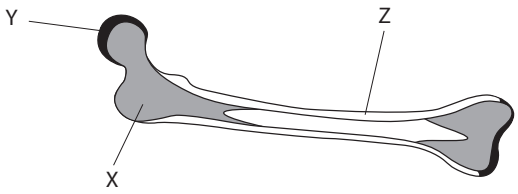
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3. Question 2 · part b.iv · 2 marks

June 2019 Paper 2R

Further practice · Attempt part b.iv only.

(b) The diagram shows a section through a bone.



(i) Identify the part labelled X.

(1)

- ☒ **A** cartilage
- ☒ **B** compact bone
- ☒ **C** ligament
- ☒ **D** spongy bone

(ii) Identify the part labelled Z.

(1)

- ☒ **A** cartilage
- ☒ **B** compact bone
- ☒ **C** ligament
- ☒ **D** spongy bone

(iii) Name the part labelled Y.

(1)

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(iv) Describe the function of the part labelled Y.

(2)

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3. Question 2 · part b.iv · 2 marks

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Further practice · Attempt part b.iv only.

(c) Describe why a balanced diet is needed for the healthy development of bone and muscle.

(6)

(Total for Question 2 = 12 marks)

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3. Question 2 · part b.iv · 2 marks · Working and answer

Working and answer

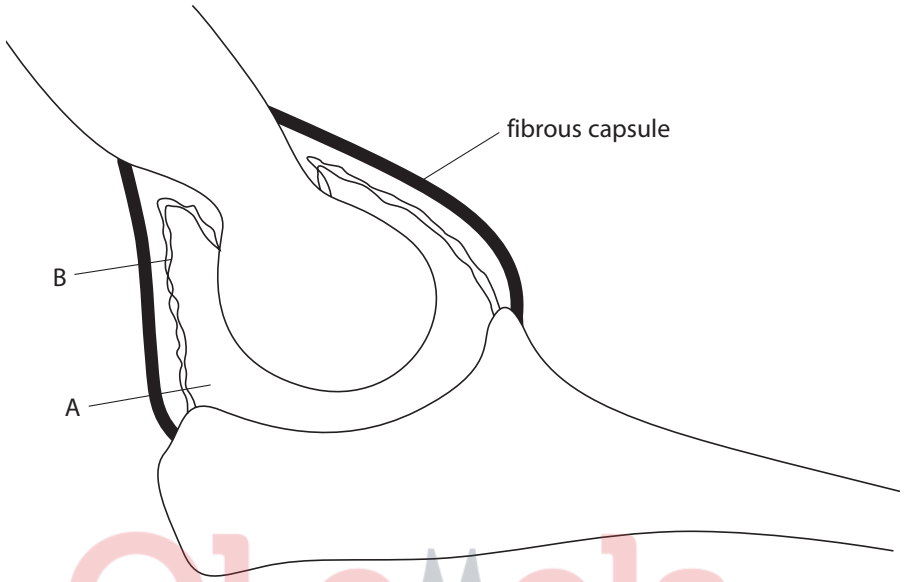
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4. Question 3 · part a.ii · 2 marks

January 2022 Paper 1

Further practice · Attempt part a.ii only.

3 The diagram shows part of an elbow joint without any cartilage.



(a) (i) Identify part A and part B. **(2)**

part A .....

part B .....

(ii) Describe the function of part A. **(2)**

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(b) (i) Complete the diagram by drawing the cartilage. **(2)**

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4. Question 3 · part a.ii · 2 marks · Working and answer

Working and answer

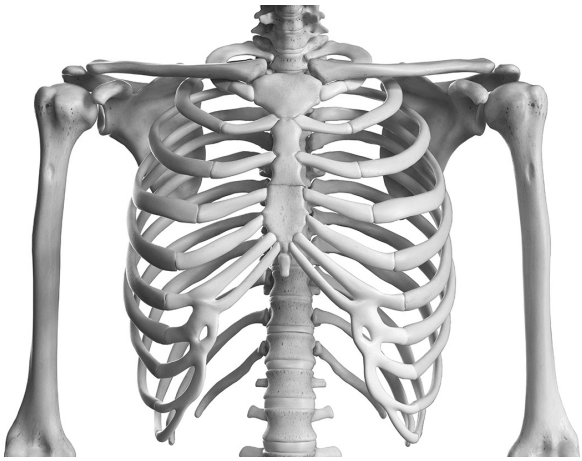
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5. Question 1 · part a.ii · 2 marks

January 2023 Paper 2

Further practice · Attempt part a.ii only.

1 (a) The image shows part of the skeleton.



(Source: © SciePro/Shutterstock)

(i) Label the clavicle, scapula and vertebral column on the image.

(3)

(ii) The vertebral column is part of the axial skeleton.

Describe the functions of the axial skeleton.

(2)

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5. Question 1 · part a.ii · 2 marks

January 2023 Paper 2

Further practice · Attempt part a.ii only.

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(b) Osteoporosis can affect the vertebral column.

Describe the symptoms of osteoporosis.

(3)

(Total for Question 1 = 8 marks)

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Working and answer

6. Question 2 · part b.iii · 2 marks

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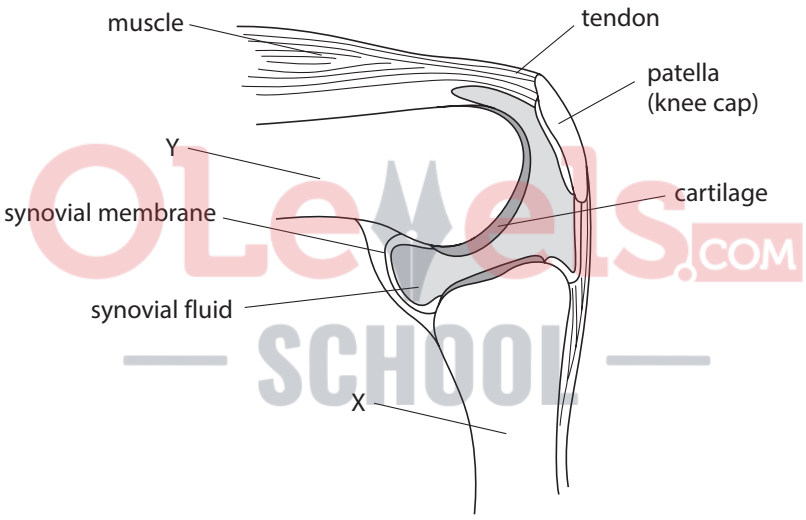
Further practice · Attempt part b.iii only.

2 The skeleton contains joints.

(a) Describe what is meant by the term **joint**.

(2)

(b) The diagram shows a knee joint.



(i) Identify bone X and bone Y.

(2)

X .....

Y .....

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6. Question 2 · part b.iii · 2 marks

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Further practice · Attempt part b.iii only.

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(ii) Give the functions of tendons and cartilage. **(4)**

tendons

cartilage

(iii) Give two structures, not shown on the diagram, that would be necessary for the joint to fully function. **(2)**

1 .....

2 .....

(Total for Question 2 = 10 marks)

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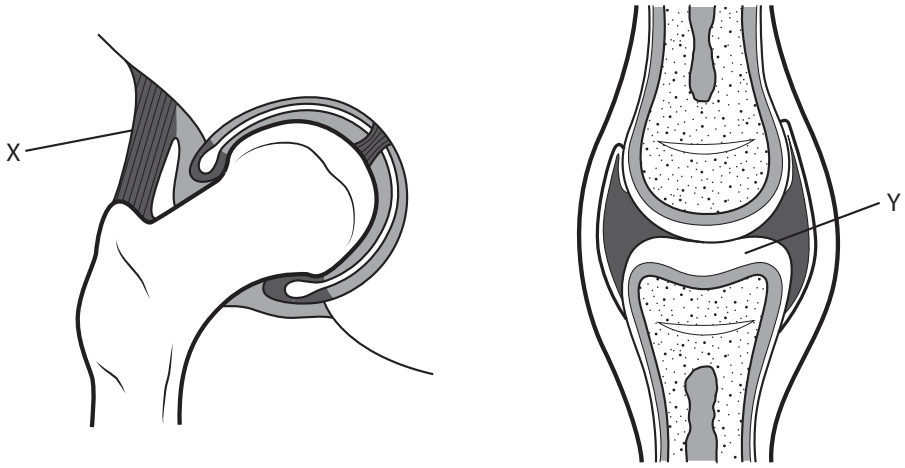
Working and answer

7. Question 5 · part b.i · 1 mark

June 2024 Paper 1R

Further practice · Attempt part b.i only.

5 The diagram shows two different types of synovial joint in the human body.



Joint A Joint B

(a) (i) Name the type of joint shown by joint A and by joint B. **(2)**

joint A

joint B

(ii) Describe the range of movement of joint A compared with the range of movement of joint B. **(2)**

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P 7 4 4 9 2 A 0 1 4 2 4

7. Question 5 · part b.i · 1 mark

June 2024 Paper 1R

Further practice · Attempt part b.i only.

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(b) Part X and part Y play an important role in the function of the joint.

(i) Give the role of part X.

(1)

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(ii) Give the role of part Y.

(1)

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(Total for Question 5 = 6 marks)

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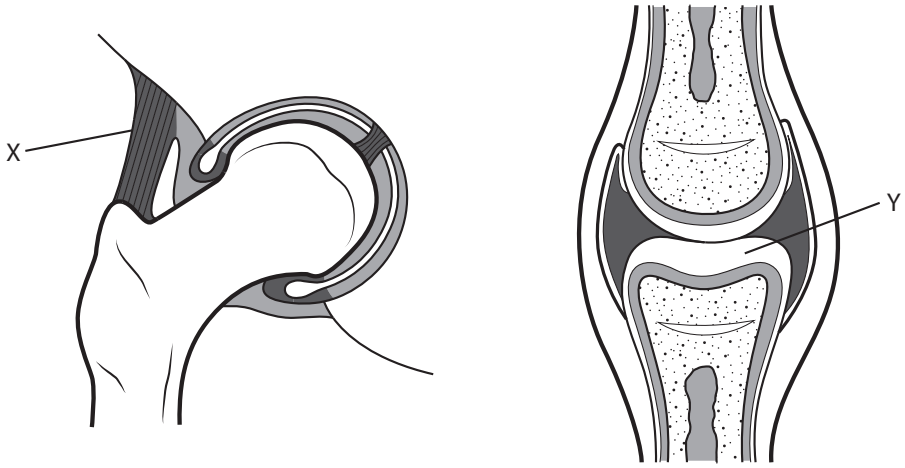
Working and answer

8. Question 5 · part b.ii · 1 mark

June 2024 Paper 1R

Further practice · Attempt part b.ii only.

5 The diagram shows two different types of synovial joint in the human body.



Joint A Joint B

(a) (i) Name the type of joint shown by joint A and by joint B. **(2)**

joint A .....

joint B .....

(ii) Describe the range of movement of joint A compared with the range of movement of joint B. **(2)**

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8. Question 5 · part b.ii · 1 mark

June 2024 Paper 1R

Further practice · Attempt part b.ii only.

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(b) Part X and part Y play an important role in the function of the joint.

(i) Give the role of part X.

(1)

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(ii) Give the role of part Y.

(1)

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(Total for Question 5 = 6 marks)

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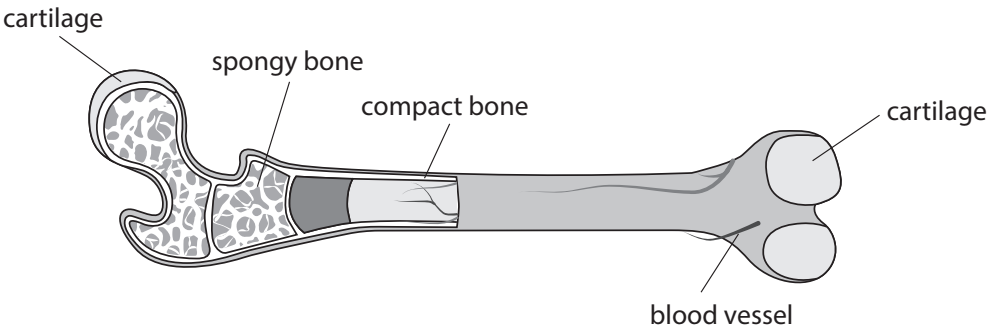
Working and answer

9. Question 1 · part a.ii · 3 marks

June 2025 Paper 2R

Further practice · Attempt part a.ii only.

1 (a) The diagram shows a bone found in the human body.



(Source: adapted from: © Designua / Shutterstock)

(i) State the name of the type of bone shown in the diagram.

(1)

.....

(ii) Place one tick (✓) in each row to give the function of each part of the bone.

(3)

Part	Function		
	reduces friction	where red blood cells are produced	provides strength
compact bone			
spongy bone			
cartilage			

(iii) Describe the function of the blood vessel in the bone.

(2)

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9. Question 1 · part a.ii · 3 marks · Working and answer

Working and answer

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10. Question 8 · part a.ii · 3 marks

June 2019 Paper 2R

Further practice · Attempt part a.ii only.

8 (a) Alkaptonuria is a condition caused by a gene mutation that produces a recessive allele.

People who have this condition are unable to fully break down two amino acids found in proteins. Instead, an intermediate compound is formed which gives a dark colour to the urine.

This intermediate compound causes damage to cartilage and heart valves.

(i) Explain how amino acids are normally metabolised.

(3)

(ii) A person with alkaptonuria produces a dark coloured urine.

Describe what other symptoms they may develop.

(3)

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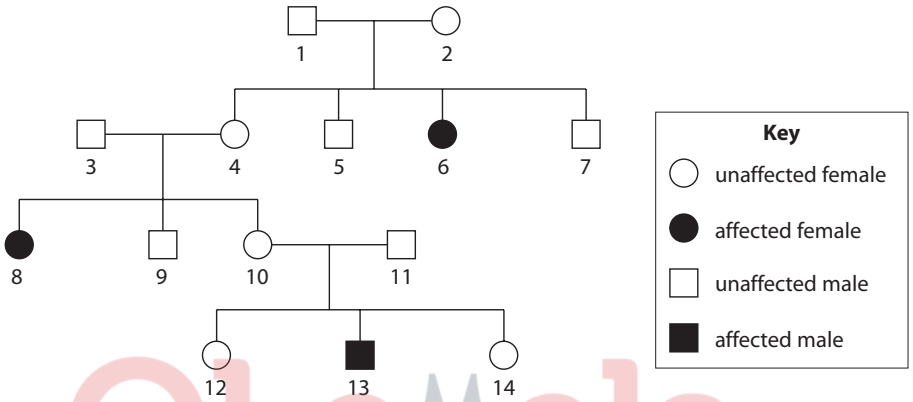
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10. Question 8 · part a.ii · 3 marks

June 2019 Paper 2R

Further practice · Attempt part a.ii only.

(b) The diagram shows the pedigree of a family affected by alkaptonuria.



(i) Use the family pedigree to explain whether the alkaptonuria condition is sex-linked.

(4)

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10. Question 8 · part a.ii · 3 marks · Working and answer

Working and answer

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11. Question 4 · part e · 2 marks

June 2019 Paper 2

Further practice · Attempt part e only.

4 Read the passage below. Use the information in the passage and your own knowledge to answer the questions that follow.

In 1916, an American doctor, Joseph Goldberger, noticed symptoms of a mystery disease that were common among his patients. These symptoms included headaches, swollen tongues, skin rashes, upset stomachs and mental illness. In one American state, North Carolina, 40% of the 30 000 people living in this area died from this mystery disease.

Goldberger investigated whether the symptoms of the mystery disease were caused by pathogens. His investigation involved injecting himself with the blood from one patient, eating the skin rash of another and swallowing fluid taken from the intestines of a third. Goldberger did not get the disease, although he did suffer from nausea and diarrhoea.

Goldberger concluded that the symptoms shown by the patients were not caused by an infectious disease. He decided that they were more likely to be caused by the poor diet of the patients.

(a) Calculate the number of people in North Carolina who died from this mystery disease (lines 4 to 5).

(2)

number of deaths = ...

(b) State the meaning of the term pathogens (line 7).

(1)

.....

.....

(c) The people were suffering from a deficiency disease called pellagra, which is caused by a lack of vitamin B in the diet.

State why deficiency diseases such as pellagra are not infectious diseases.

(1)

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11. Question 4 · part e · 2 marks

June 2019 Paper 2

Further practice · Attempt part e only.

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(d) Describe a method that Goldberger could use to show that the symptoms of his patients were caused by a lack of vitamin B in their diet.

(3)

(e) Scurvy is another deficiency disease.

Scurvy affects the production of a protein called collagen.

Collagen is a component of many body tissues including cartilage.

Explain why a person with scurvy often experiences pain when walking.

(2)

(Total for Question 4 = 9 marks)

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11. Question 4 · part e · 2 marks · Working and answer

Working and answer

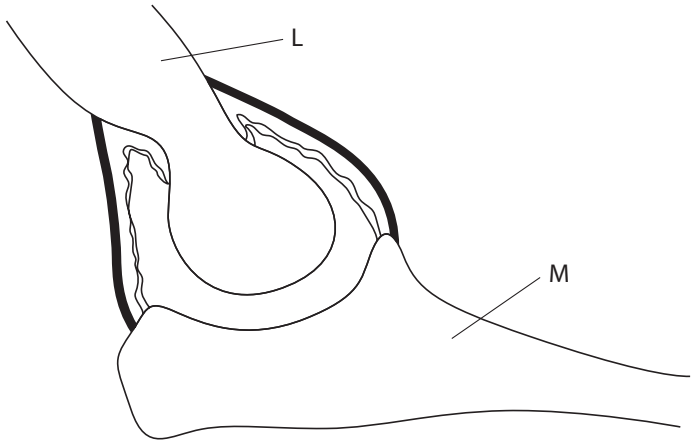
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12. Question 6 · part c · 2 marks

November 2021 Paper 2

Further practice · Attempt part c only.

6 The diagram shows some of the structures of the elbow joint.



(a) State what is meant by the term **joint**.

(2)

.....

.....

.....

.....

(b) (i) Give the name of bone L and the name of bone M.

(2)

L .....

M .....

(ii) Cartilage has not been shown on the diagram.

Draw the missing cartilage on the diagram.

(2)

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12. Question 6 · part c · 2 marks

November 2021 Paper 2

Further practice · Attempt part c only.

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(c) Explain the effects on a person if the cartilage in this joint is missing or damaged.

(2)

(d) Tendons are important in the functioning of a joint.

Explain the importance of tendons in the movement of bone M.

(4)

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(Total for Question 6 = 12 marks)

12. Question 6 · part c · 2 marks · Working and answer

Working and answer

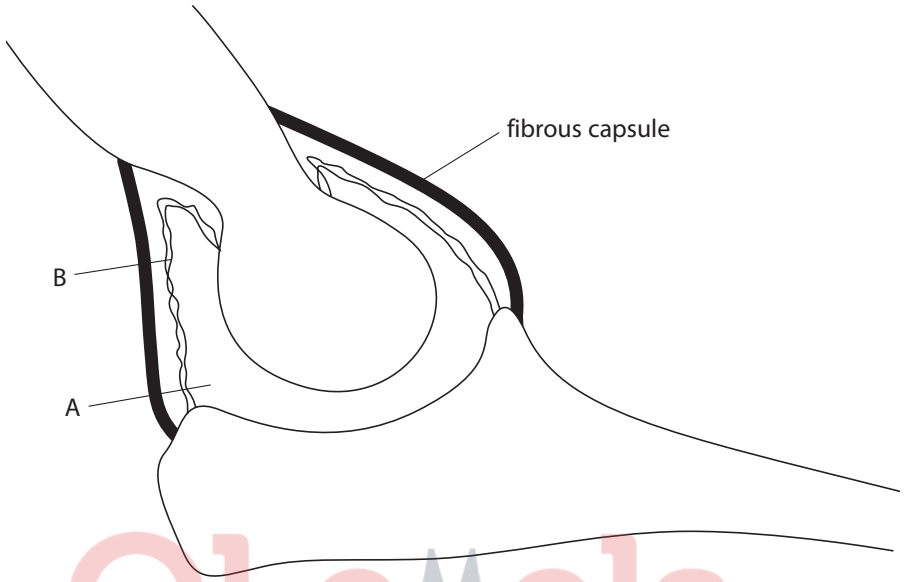
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13. Question 3 · part b.ii · 2 marks

January 2022 Paper 1

Further practice · Attempt part b.ii only.

3 The diagram shows part of an elbow joint without any cartilage.



(a) (i) Identify part A and part B. **(2)**

part A .....

part B .....

(ii) Describe the function of part A. **(2)**

.....

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(b) (i) Complete the diagram by drawing the cartilage. **(2)**

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13. Question 3 · part b.ii · 2 marks

January 2022 Paper 1

Further practice · Attempt part b.ii only.

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(ii) Cartilage often breaks down as people become older.

Explain the effects this has on the working of the joint.

(2)

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(c) Explain the role of tendons in the movement of the lower arm.

(5)

(Total for Question 3 = 13 marks)

13. Question 3 · part b.ii · 2 marks · Working and answer

Working and answer

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